KitabHoshmand

AI-Powered PDF
Study Companion

Software Engineering
Monograph

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Supervisor: [Supervisor Name] · 2026



PRIVATE LIBRARY · SEARCH · VOICE · CHAT · STUDY GUIDES

A static PDF becomes an interactive study resource

Academic material often arrives as long PDF files.

KitabHoshmand creates one private workflow for reading, searching, discussing, and reviewing that material.

Designed for authenticated individuals studying their own documents

PDF

- Searchable text segments
- Voice and text conversation
- Saved study guides

The project addresses four practical study gaps

01

Passive reading

Basic viewers do not support dialogue with the document.

02

Fragmented tools

Reading, notes, search, and summaries are often separated.

03

Weak continuity

Questions and explanations are difficult to revisit later.

04

Limited interaction

Long documents are hard to explore through natural conversation.

Objectives align directly with implemented behavior

General objective

Build a secure, interactive study system around user-selected PDF documents.

- Ingest and search PDF content
- Support voice and text study
- Preserve sessions and transcripts
- Generate grounded study guides
- Isolate each user’s data

One workflow connects upload, study, and review

01

Sign in

Access a private account

02

Upload

Select a searchable PDF and voice

03

Process

Extract, segment, validate, and store

04

Study

Voice, chat, search, or summarize

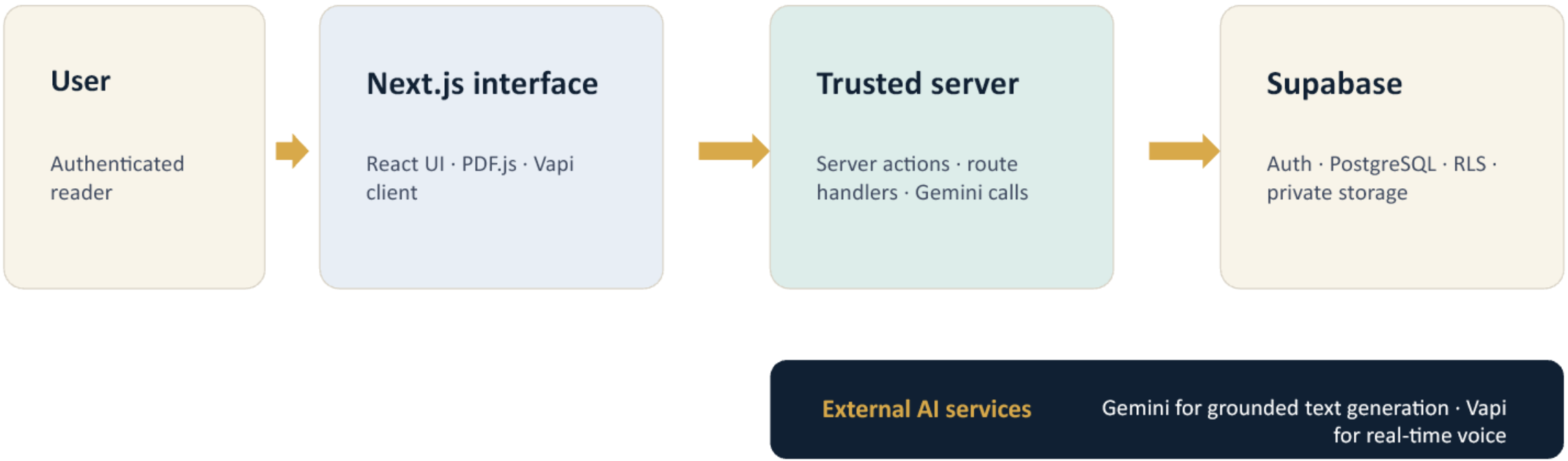
05

Revisit

Open saved sessions and guides

Every study mode remains attached to one owned book.

The architecture separates interface, server logic, and services



Each technology has a specific responsibility

Main technologies

Next.js + React	Application interface, routes, server components, and server actions
TypeScript	Typed application logic and shared data contracts
Supabase	Authentication, PostgreSQL, RLS, RPCs, and private storage
Tailwind CSS	Responsive styling and visual system

Supporting tools

PDF.js	Text extraction and cover rendering
Zod + React Hook Form	Validation and form state
Gemini	Grounded chat and study guides
Vapi + ElevenLabs voices	Real-time voice interaction
Vitest + Playwright	Automated verification

Six implemented capabilities form the working system

01

Account access

Sign-up, sign-in, recovery, and reset

02

PDF ingestion

Validation, extraction, segmentation, and storage

03

Private library

Owned book listing and title/author search

04

Voice study

Live sessions with saved final transcripts

05

Grounded chat

Book-segment retrieval and persisted messages

06

Study guides

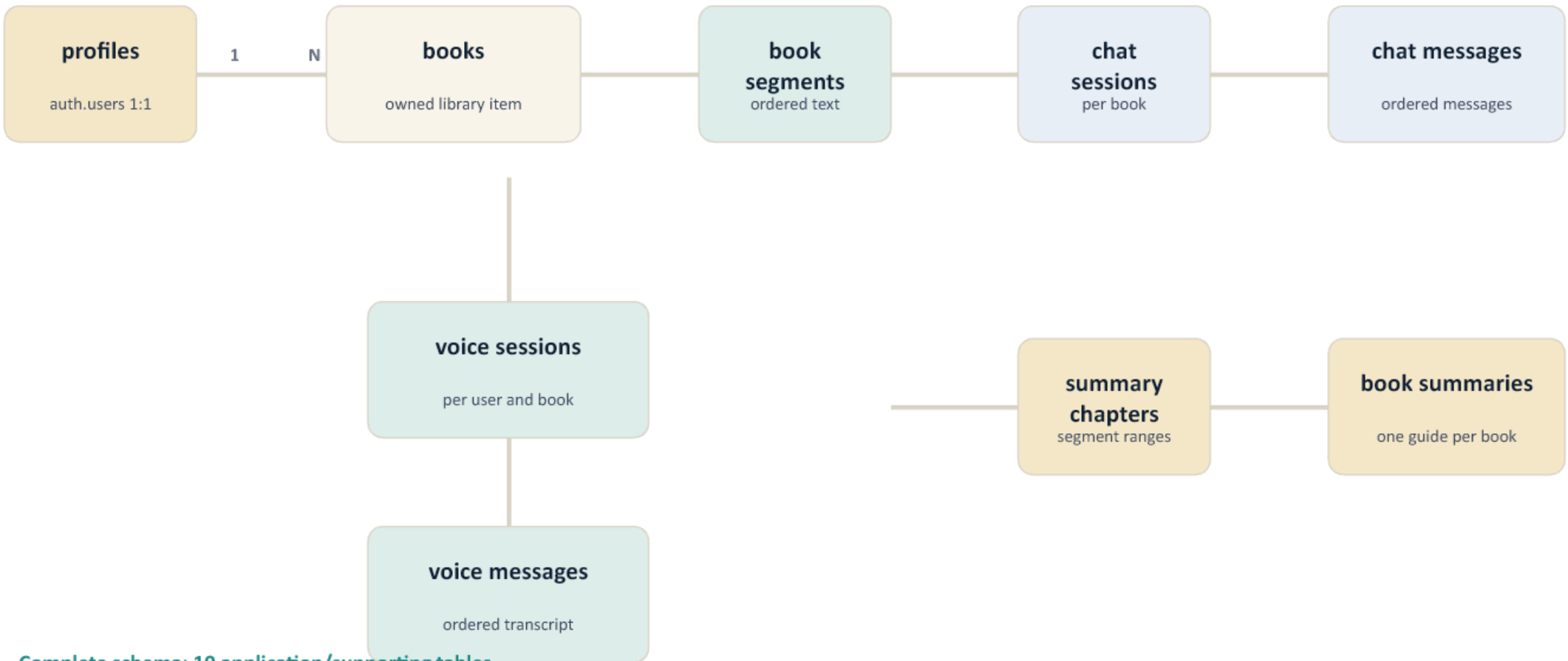
Chapter summaries, concepts, takeaways, and questions

Upload is the core end-to-end workflow

1	**Validate**	Zod checks metadata, MIME types, and size limits.
2	**Extract**	PDF.js reads text and creates 500-word overlapping segments.
3	**Upload**	PDF and cover enter a private user/book storage folder.
4	**Revalidate**	The server checks UUIDs, paths, size, and segment order.
5	**Persist**	The ingest_book RPC inserts the book and its segments.

Failure handling: uploaded objects are removed if ingestion does not complete.

The database follows ownership and study activity



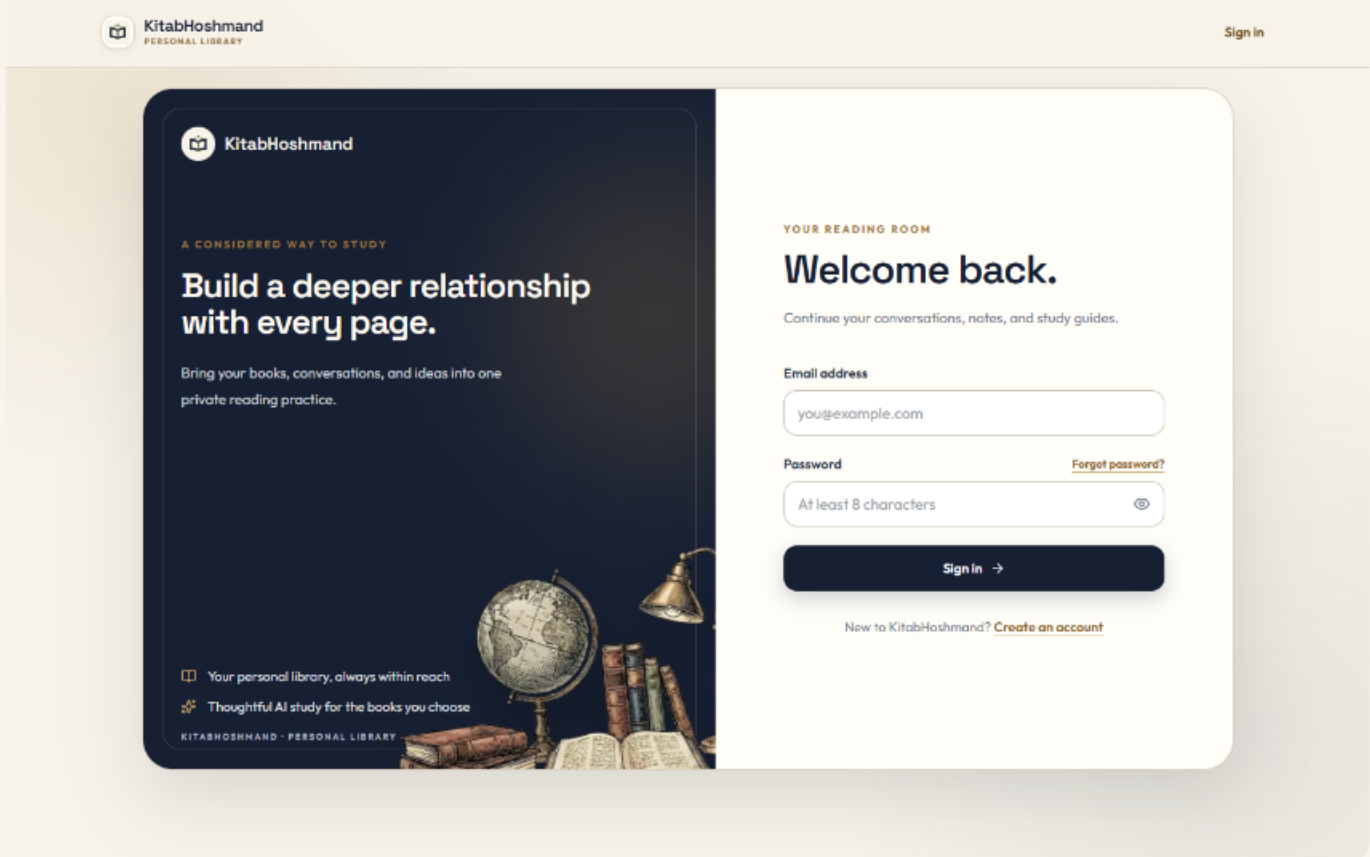
Complete schema: 10 application/supporting tables

Protection is applied in layers

1 Authentication	Supabase email/password sessions and verified current-user checks
2 Application authorization	Protected pages/actions call requireUser()
3 Input validation	Zod schemas, path checks, size limits, and message limits
4 Database protection	Constraints, ownership checks, RPC validation, and Row Level Security
5 Private file access	User-ID storage folders and short-lived signed cover URLs

Vapi tool calls additionally require a timing-safe bearer-secret check.

The implemented interface emphasizes private study



Actual running route

/sign-in

- Email/password access
- Password recovery
- Account creation
- No credentials displayed

Current verification confirms core logic and build readiness

10 / 10

Vitest tests passed

4 test files · segmentation, Gemini, summaries

PASS

TypeScript check

npx tsc --noEmit · exit code 0

PASS

Production build

Next.js compiled and generated all routes

Coverage boundary

Live Supabase policies, authenticated upload, Vapi integration, and complete browser workflows remain outside the executed automated checks.

Limitations define a realistic next iteration

Current limitations

- No OCR for scanned PDFs
- External Vapi assistant setup
- No admin roles, analytics, or export
- Limited live integration coverage
- No verified production deployment

Appropriate improvements

- OCR for image-only documents
- End-to-end Supabase and voice tests
- Book-scoped Vapi authorization
- Exportable study materials
- Accessibility and usability evaluation

KitabHoshmand demonstrates a complete private study workflow.

A user-selected PDF becomes searchable, conversational, and reusable study material through document processing, managed data services, and layered access control.

Thank You

Questions & Discussion

KITABHOSHMAND